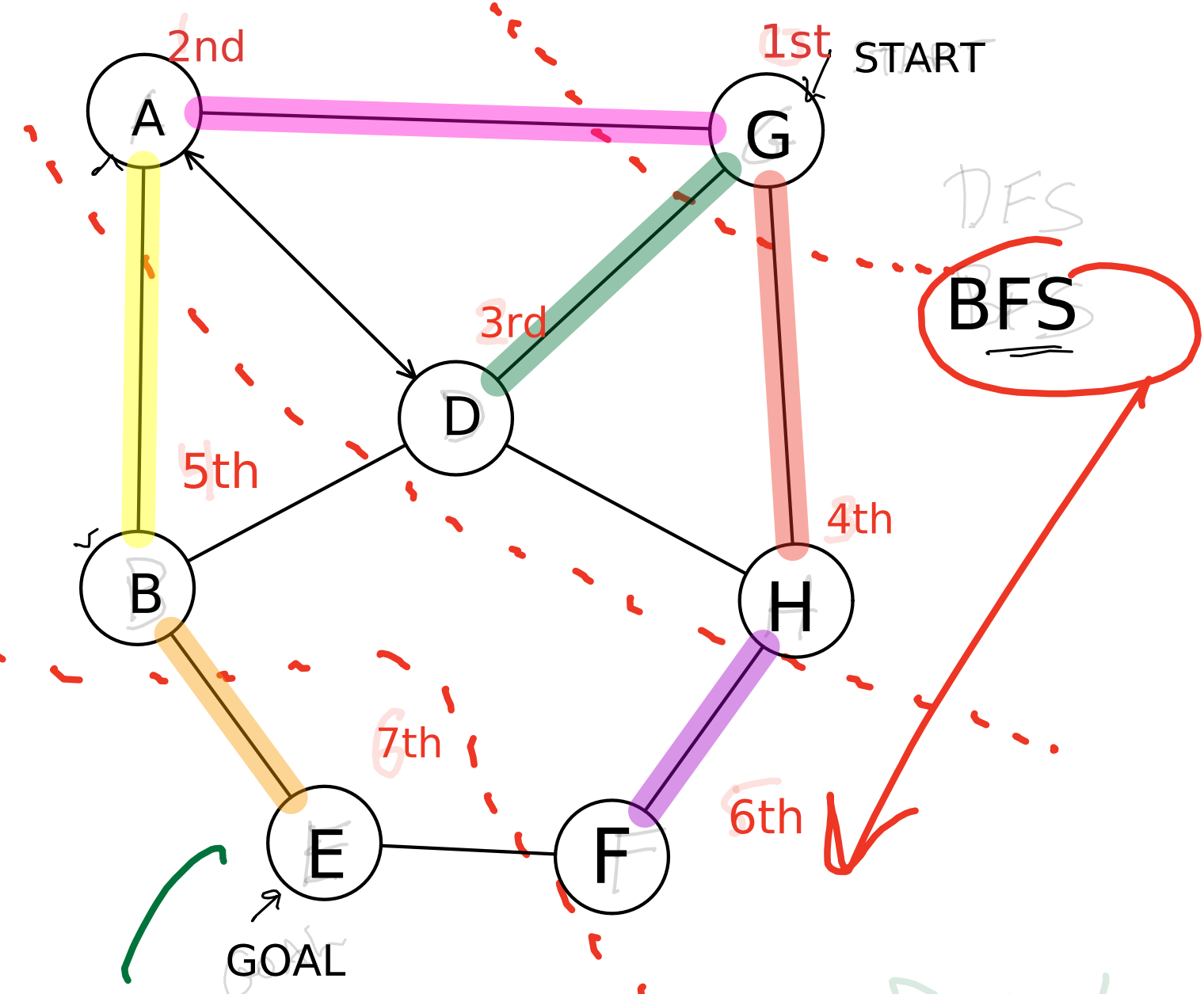


visit a node for the first time:

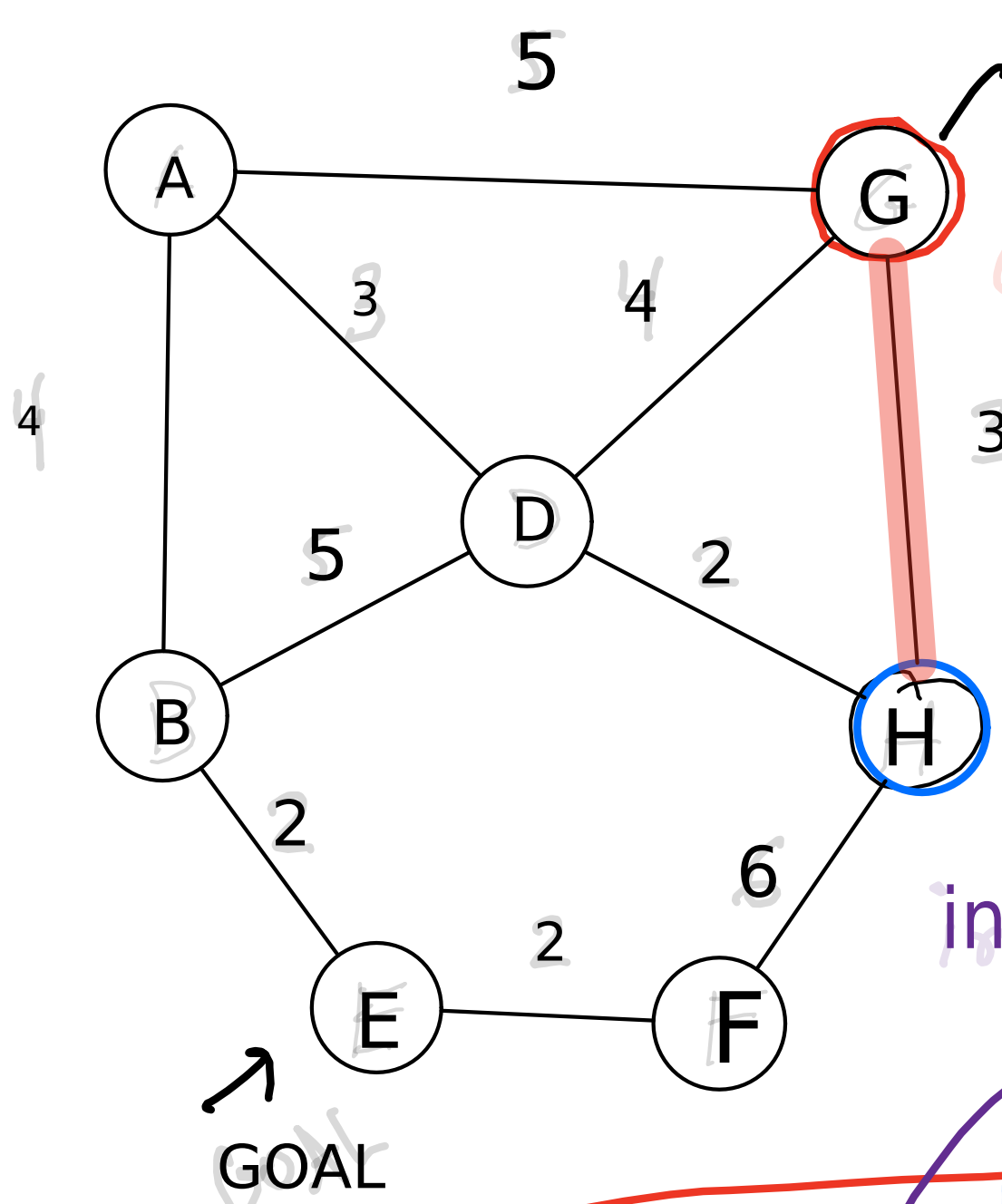
① add it to tree

② Invariant per iteration

"shortest path tree"

What if we have edge weights?

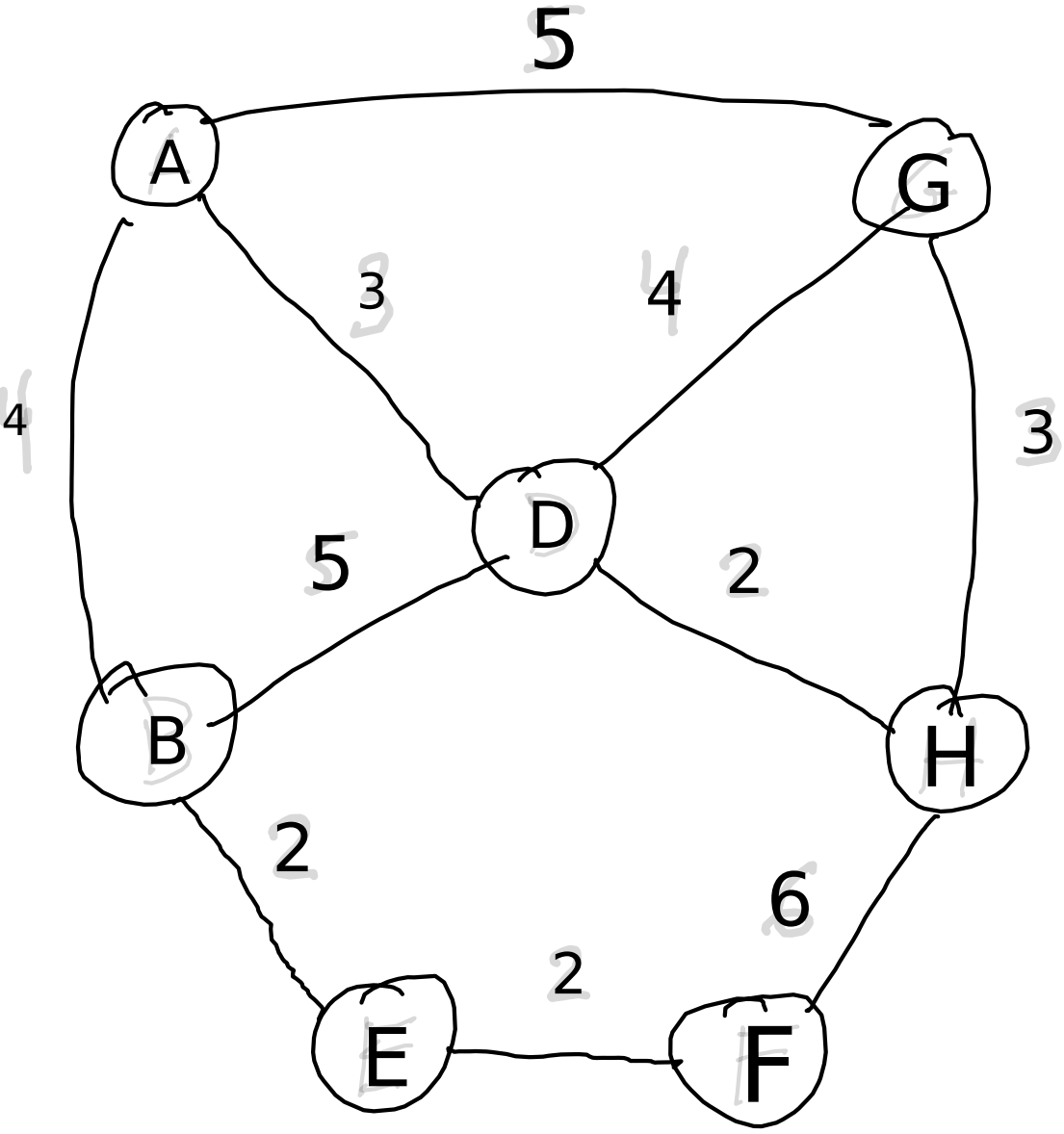
Guarantee: all weights are non-negative

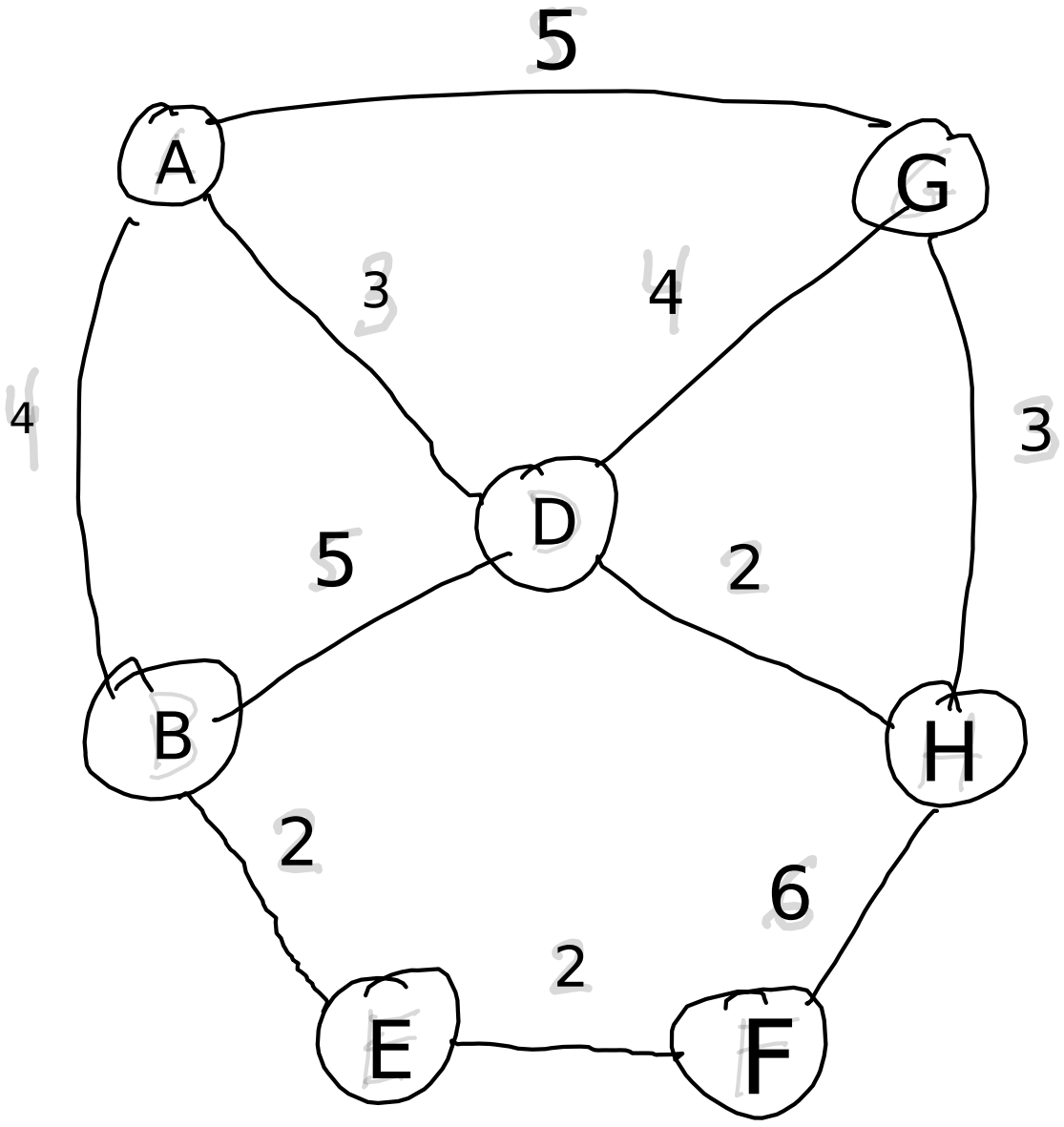


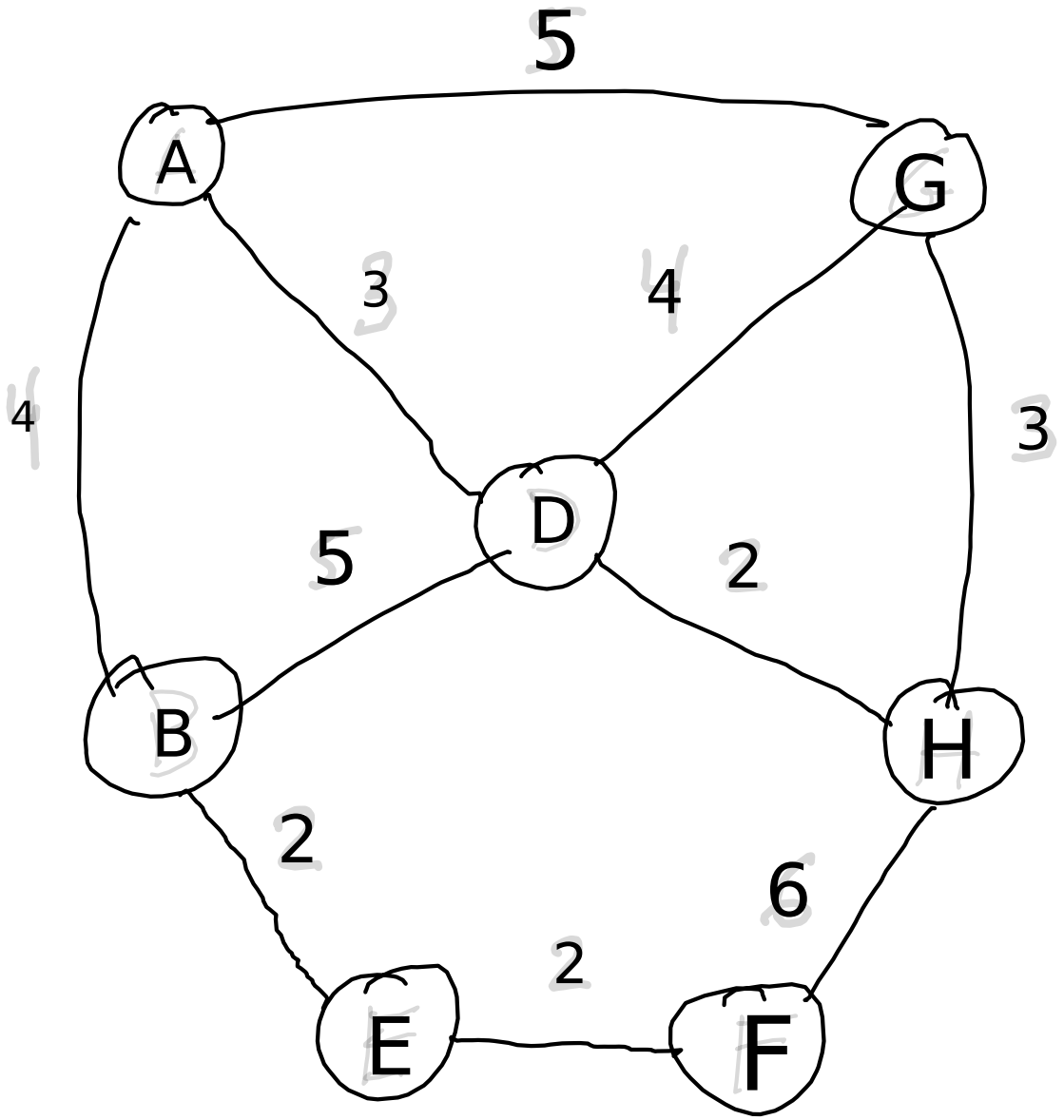
in total cost
root → node

BFS Invar: tree is a shortest-
→ path tree (in #hops)
BFS

Update this
invariant to
handle weights


5
A
G
3
4
4
3
D
5
2
B
H
2
6
2
E
F


5
A
G
3
4
4
3
D
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2
B
H
2
6
2
E
F


5
A
G
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3
D
5
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B
H
2
6
2
E
F